Torsion from Wilson-line lattice shifts in heterotic orbifolds:
a tested phase formula, its selection rule, and a Scherk–Schwarz window

W(3,3) programme — Claude track (Passes 12265–12333)

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Abstract

Shifting a Wilson line of a heterotic orbifold by a lattice vector λ changes no consistency condition but can change the spectrum. We write the change of every projection phase as an alternating bicharacter $B(g,h)$ on commuting space-group elements and test it against the orbifolder engines: $B = 0$ on all commuting pairs implies an unchanged spectrum (232/232), every change has $B \neq 0$ (61/61), and a pre-registered prediction for fresh $T^6/\mathbb{Z}_4$ models is correct in 124/124 cases. Brother models and generalized discrete torsion are known (Plöger, Ramos-Sánchez, Ratz, Vaudrevange); what is new here is the explicit, tested formula, its use on the $W(3,3)$ orbifold routes, and a radius analysis of a Scherk–Schwarz brother with three chiral generations.

1 The phase formula

Let $\lambda \in \Gamma_{E_8\times E_8}$ be added to the Wilson lines in a set S of lattice directions, and write $\sigma_g = \sum_{x\in S} m_x(g)$ for the winding labels of a space-group element g along S. In a sector h the left-moving momenta form the coset $\Gamma + V_h$, which the shift does not change.

Lemma 1. *The projection phase of g on sector h changes by*

$$B(g,h) = \tfrac{1}{2}\big(\sigma_g\, V_h\cdot\lambda - \sigma_h\, V_g\cdot\lambda\big) \pmod 1,$$

an alternating bicharacter on commuting pairs ($\lambda^2 \in 2\mathbb{Z}$ makes the quadratic term integral).

Tests. On 320 relabellings ($T^6/\mathbb{Z}_3$ and $T^6/\mathbb{Z}_2\times\mathbb{Z}_2$), B evaluated on every commuting pair predicts every spectrum change; $B \neq 0$ is necessary, not sufficient (three cases with $B \neq 0$ leave every massless state unchanged). A first version that dropped fixed-point labels mispredicted 16/100 $T^6/\mathbb{Z}_4$ brothers; with them, predictions written and pushed before the engine run were correct in 124/124 cases.

2 Consequences

Proposition 1. *On $T^6/\mathbb{Z}_3$ a lattice shift preserving the Plöger conditions induces no torsion: every twisted centralizer is generated by the sector element and the central fermion-parity element, on which B vanishes by antisymmetry and by the conditions themselves.*

This explains why the non-supersymmetric orbifolder, which skips two of the conditions (a verbose-only failure flag), still produced correct $T^6/\mathbb{Z}_3$ spectra: all 497 affected corpus inputs have a lattice-shift repair with an identical spectrum (983/983). Torsion requires commuting elements whose effective orders share a factor, and it also enters through fixed-point labels.

- Supersymmetric $\mathbb{Z}_6$-II Standard Models: 1471/1471 brothers identical.
- $\mathbb{Z}_2 \times \mathbb{Z}_6$-I: 12 further three-generation supersymmetric Standard Models. All fail later tests: renormalisable QLd^c, or textures.
- Non-supersymmetric $\mathbb{Z}_6$-II: the fermion-parity element and an order-2 translation give a Scherk–Schwarz bit.

3 A Scherk–Schwarz brother with three chiral generations

Among all 1399 non-supersymmetric $\mathbb{Z}_6$-II models of the route, exactly one lineage (15 variants of one parent) has a Scherk–Schwarz brother with three chiral left-right generations and no chiral fractional charge. Its brother has no tachyon at zero momentum and winding (engine solver). The torsion shifts the momentum of the fermion-parity-twisted states by 1/2 for every winding (a lattice Wilson line is not a freely acting translation). Hence:

- it is tachyonic for $R_6 > \sqrt{3\alpha'/8}$;
- exact $E_8 \times E_8$ level minima ($-1/3$ against $-2/3$ needed, 0 against $-1/2$) close every small-radius channel;
- a winding channel along a second circle requires $R_5 \gtrsim \sqrt{2\alpha'}$.

Computed: zero-winding spectra, right-moving tachyonic vacua, the momentum shift, the level minima. *Argued:* the circle mass formula; the projections of winding states are not computed.

Obstructions. The hypercharge normalisation is $k_Y = 8/3$. One-loop running with the model's full light spectrum above any common threshold between 10^3 and 10^{16} GeV needs $k_Y \in [0.21, 1.28]$ at a perturbative $\alpha_2 = \alpha_3$ crossing. The one-loop potential in the string-scale window needs the states the momentum shift moves to half-integer momentum, which the engine does not produce. The candidate is therefore not a Standard Model.

Record of corrections. The radius statement was revised twice before reaching the form above:

- winding instead of momentum;
- an odd-winding flip that holds only for free actions.

A suspicion that the engine's "excited tachyons" were massless was withdrawn after its solver was run verbosely. Each correction is recorded in the producers of Passes 12299–12327.

Reproducibility. Producers `w33_pass12265…` through `w33_pass12328…` in `analysis/`; the pre-registration is commit `e570f4b1f`.